Database & Screens — Completed Implementation

> *AI-Marketing-New Platform Source: Verified from `backend/Models.cs`, `backend/AppDbContext.cs`, `backend/Program.cs`, `frontend/src/main.js`*

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1. Screens Completed (Frontend)

All screens are implemented as render functions inside `frontend/src/main.js`. The router dispatches via `renderScreen(screenId)`.

1.1 Screen Inventory (36 Screens)

#	Screen ID	Render Function	Role	Status
1	`Login`	`renderLoginScreen()`	All	✅ Done
2	`Dashboard`	`renderDashboardScreen()`	All	✅ Done
3	`CampaignBuilder`	`renderCampaignBuilderWizard(step, campaignId)`	Expert	✅ Done
4	`Objective`	`renderObjectiveScreen()`	Expert	✅ Done
5	`Targeting`	`renderTargetingScreen()`	Expert	✅ Done
6	`Research`	`renderStrategyHub()`	Expert	✅ Done
7	`CreativeConfig`	`renderCreativeConfigScreen()`	Expert	✅ Done
8	`Studio`	`renderStudioScreen()`	Expert	✅ Done
9	`Monitoring`	`renderMonitoringScreen()`	CMO / PPP	✅ Done
10	`Budget`	`renderBudgetScreen()`	CMO	✅ Done

11	`ApprovedAssets`	`renderApprovedAssetsScreen()`	PPP	✅ Done
12	`Config`	`renderConfigScreen()`	Admin	✅ Done
13	`RoleManagement`	`renderRoleManagementScreen()`	Admin	✅ Done
14	`UserManagement`	`renderUserManagementScreen()`	Admin	✅ Done
15	`CompanyProfile`	`renderCompanyProfileScreen()`	Admin	✅ Done
16	`Calendar`	`renderCalendarScreen()`	Admin	✅ Done
17	`Guideline`	`renderGuidelineScreen()`	Admin	✅ Done
18	`Assets`	`renderAssetsScreen()`	Expert / Admin	✅ Done
19	`BudgetMatrix`	`renderBudgetMatrixScreen()`	CMO	✅ Done
20	`Approvals`	`renderRedesignedApprovalsScreen()`	CMO	✅ Done
21	`Notifications`	`renderNotificationsScreen()`	All	✅ Done
22	`DeploySelection`	`renderEnhancedDeploySelectionScreen()`	PPP	✅ Done
23	`AdPerformance`	`renderRealAdPerformanceScreen()`	CMO / PPP	✅ Done
24	`GlobalDashboard`	`renderGlobalDashboard()`	Super Admin	✅ Done
25	`CompanyManagement`	`renderCompanyManagementScreen()`	Super Admin	✅ Done
26	`AuditLog`	`renderAuditLogScreen()`	Super Admin	✅ Done
27	`SystemConfig`	`renderSystemConfigScreen()`	Super Admin	✅ Done
28	`AdAccountManagement`	`renderAdAccountManagementScreen()`	Admin	✅ Done
29	`BillingSettings`	`renderBillingSettingsScreen()`	Admin	✅ Done

30	`CampaignReports`	`renderCampaignReportsScreen()`	CMO	✅ Done
31	`CrossPlatformAnalytics`	`renderCrossPlatformAnalyticsScreen()`	CMO	✅ Done
32	`DeploymentHistory`	`renderDeploymentHistoryScreen()`	PPP	✅ Done
33	`ABTestResults`	`renderABTestResultsScreen()`	PPP / CMO	✅ Done
34	`AudienceInsights`	`renderAudienceInsightsScreen()`	Expert	✅ Done
35	`CompetitorResearch`	`renderCompetitorResearchScreen()`	Expert	✅ Done
36	*(onboarding)*	`renderOnboardingWizard()`	Admin	✅ Done

1.2 Screen Details by Role

Super Admin Screens (4)

Screen	Description	Backend API Used
GlobalDashboard	Platform-wide stats: total companies, users, campaigns, revenue	`GET /api/super-admin/dashboard`
CompanyManagement	List, create, edit, suspend companies; view per-company users & campaigns	`GET/POST/PUT/DELETE /api/super-admin/companies`
AuditLog	Full activity log across all tenants, filterable by company	`GET /api/super-admin/audit-log`
SystemConfig	System-level configuration (platform settings)	Local state

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Admin Screens (7)

Screen	Description	Backend API Used
Config (Platform Config)	Manage Facebook/TikTok API keys, Gemini API key	`GET/PUT /api/company-settings`
AdAccountManagement	Add/edit/delete ad platform accounts (tokens, account IDs, page IDs)	`GET/POST/PUT/DELETE /api/ad-accounts`
UserManagement	View users, revoke/activate/permanently delete; manage invitations	`GET /api/rbac/users`, `DELETE/PATCH /api/rbac/users/{id}`, `POST /api/invitations`

RoleManagement	Create/edit/delete roles; assign screen permissions per role; user-level overrides	`GET/POST/PUT/DELETE /api/rbac/roles`, `GET/POST /api/rbac/role-permissions`, `GET/POST /api/rbac/user-permissions`
CompanyProfile	Edit company name, logo, industry, website, contact details	Local/settings
Calendar	Global operations calendar with campaign events	Local state
Guideline (Brand DNA)	Configure brand tone, language, palette, typography, whitelist/blacklist	`GET/POST /api/guidelines`
BillingSettings	Subscription plan and billing configuration	Local state

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Marketing Expert Screens (8)

Screen	Description	Backend API Used
Objective	Select campaign objective (Awareness / Traffic / Engagement / Conversions)	`GET /api/campaign-objectives`
Targeting	Define target audience: country, language, age range, gender; multiple target sets	Local state → campaign
Research (Strategy Hub)	3-phase AI strategy: brief → L1 questions → L2 questions via Gemini	`POST /api/gemini/questions`, `POST /api/gemini/follow-up`
CreativeConfig (Creative Hub)	Define platform, aspect ratio, image/video counts, video duration	`POST /api/campaigns`
Studio (Creative Studio)	Review generated assets, edit copy, approve/discard, submit to PPP queue	`GET /api/assets`, `POST /api/assets/approve`, `POST /api/cmo/queue`
Assets	Browse all working assets in `/Assets` folder	`GET /api/assets`
AudienceInsights	Audience analysis and insights dashboard	`GET /api/analytics/overview`
CompetitorResearch	Competitor research tools	Local state
CampaignBuilder	Multi-step campaign creation wizard	`POST /api/campaigns`, workflow APIs

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CMO (Business Admin) Screens (6)

Screen	Description	Backend API Used
BudgetMatrix (Budget & Matrix)	Set total budget, platform allocations, weight sliders; define allowed platforms	`GET/POST /api/budgets`

Approvals (CMO Queue)	Review PPP-approved assets; final approve/reject each ad with notes	`GET/POST /api/cmo/queue`
Monitoring (AI Monitoring)	Active ads, total spend, efficiency metrics, auto-action recommendations	`GET /api/analytics/overview`
AdPerformance	Real performance metrics: impressions, CTR, CPC, ROAS, conversions	`GET /api/analytics/campaigns/{id}/metrics`
CampaignReports	Campaign-level reporting with platform breakdown	`GET /api/analytics/overview`, `GET /api/analytics/top-performers`
CrossPlatformAnalytics	Cross-platform comparison analytics	`GET /api/analytics/platforms`
Budget	Budget overview and spend tracking	`GET /api/budgets/current`
ABTestResults	View A/B test results with statistical significance	`GET /api/ab-tests/{id}/results`

PPP Specialist Screens (5)

Screen	Description	Backend API Used
ApprovedAssets (PPP Queue)	Review submitted assets; enter per-ad budget; approve/reject	`GET/POST /api/ppp/queue`
DeploySelection	Select CMO-approved ads, choose platform, configure deployment	`GET /api/ppp/queue`
Deploy (via DeploySelection)	Execute Facebook / TikTok deployment with full API orchestration	`POST /api/deploy/facebook`, `POST /api/deploy/tiktok-adgroup`, `POST /api/deploy/unified`
AdPerformance	Ad performance monitoring post-deployment	`GET /api/analytics/campaigns/{id}/metrics`
DeploymentHistory	Full history of all deployment actions and logs	`GET /api/deployment-logs`
Monitoring	AI monitoring dashboard (shared with CMO)	`GET /api/analytics/overview`

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2. Database Tables — Complete Reference

Total Tables Implemented: 29 (28 planned + `user_screens` for user-level permission overrides)

DOMAIN 1 — Multi-Tenancy (3 Tables)

Table 1: `companies`

C# Model: `Company` | **Purpose:** Core multi-tenancy — every company's data is isolated

Column	Type	Constraints	Notes
`id`	INT	PK, auto-increment	
`name`	VARCHAR(200)	NOT NULL	
`slug`	VARCHAR(100)	NOT NULL, UNIQUE	URL-friendly identifier
`industry`	VARCHAR(100)	NULLABLE	
`website`	VARCHAR(500)	NULLABLE	
`logo_url`	VARCHAR(500)	NULLABLE	
`email`	VARCHAR(200)	NULLABLE	
`phone`	VARCHAR(50)	NULLABLE	
`address`	TEXT	NULLABLE	
`country`	VARCHAR(100)	NULLABLE	
`timezone`	VARCHAR(50)	DEFAULT 'UTC'	
`currency`	VARCHAR(10)	DEFAULT 'USD'	
`status`	VARCHAR(20)	DEFAULT 'active'	active / suspended / archived
`subscription_plan`	VARCHAR(50)	DEFAULT 'free'	free / starter / pro / enterprise
`max_users`	INT	DEFAULT 10	
`max_campaigns`	INT	DEFAULT 50	
`created_by`	INT	NULLABLE	FK → users.id (Super Admin)
`created_at`	TIMESTAMP	DEFAULT NOW()	
`updated_at`	TIMESTAMP	DEFAULT NOW()	

Indexes: `uq_companies_slug` (UNIQUE) **Navigation:** → CompanySetting (1:1), → Users, Roles, AdAccounts, Campaigns

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Table 2: `company_settings`

C# Model: `CompanySetting` | **Purpose:** Per-company configuration

Column	Type	Constraints	Notes
`id`	INT	PK	

`company_id`	INT	FK → companies.id, UNIQUE	1:1 with company
`default_language`	VARCHAR(10)	DEFAULT 'en'	
`notification_email`	VARCHAR(200)	NULLABLE	
`max_daily_budget`	DECIMAL	NULLABLE	
`auto_approve_below`	DECIMAL	NULLABLE	auto-approve threshold
`require_cmo_approval`	BOOL	DEFAULT true	
`require_brand_check`	BOOL	DEFAULT false	
`default_bid_strategy`	VARCHAR(50)	DEFAULT 'lowest_cost'	
`default_platforms`	TEXT[]	DEFAULT '{facebook}'	
`gemini_api_key`	VARCHAR(500)	NULLABLE	encrypted
`created_at`	TIMESTAMP	DEFAULT NOW()	
`updated_at`	TIMESTAMP	DEFAULT NOW()	

Indexes: `uq_settings_company` (UNIQUE on company_id)

Table 3: **`company_ad_accounts`**

C# Model: `CompanyAdAccount` | **Purpose:** Per-company ad platform credentials (replaces hardcoded appsettings.json)

Column	Type	Constraints	Notes
`id`	INT	PK	
`company_id`	INT	FK → companies.id	
`platform`	VARCHAR(50)	NOT NULL	facebook / tiktok / youtube / google_ads
`account_name`	VARCHAR(200)	NULLABLE	friendly name
`account_id`	VARCHAR(200)	NOT NULL	e.g. act_2537893049860881
`access_token`	TEXT	NOT NULL	encrypted
`refresh_token`	TEXT	NULLABLE	encrypted
`token_expires_at`	TIMESTAMP	NULLABLE	
`page_id`	VARCHAR(200)	NULLABLE	Facebook page
`pixel_id`	VARCHAR(200)	NULLABLE	tracking pixel
`developer_token`	VARCHAR(200)	NULLABLE	Google Ads
`customer_id`	VARCHAR(200)	NULLABLE	Google Ads
`status`	VARCHAR(20)	DEFAULT 'active'	active / expired / revoked / error

`last_tested_at`	TIMESTAMP	NULLABLE	
`last_error`	TEXT	NULLABLE	
`created_at`	TIMESTAMP	DEFAULT NOW()	
`updated_at`	TIMESTAMP	DEFAULT NOW()	

Indexes: `idx_adaccounts_company`, `uq_adaccounts_platform` (company_id + platform + account_id UNIQUE)

DOMAIN 2 — Auth & RBAC (6 Tables)

Table 4: `users`

C# Model: `User` | **Purpose:** User accounts and credentials

Column	Type	Constraints	Notes
`id`	INT	PK	
`username`	VARCHAR(50)	NULLABLE	
`password_hash`	TEXT	NULLABLE	BCrypt hash
`email`	VARCHAR(200)	NULLABLE, UNIQUE	
`first_name`	VARCHAR(100)	NULLABLE	
`last_name`	VARCHAR(100)	NULLABLE	
`avatar_url`	VARCHAR(500)	NULLABLE	
`phone`	VARCHAR(50)	NULLABLE	
`role_id`	INT	FK → roles.id	
`company_id`	INT	FK → companies.id, NULLABLE	NULL = Super Admin
`is_super_admin`	BOOL	DEFAULT false	
`status`	VARCHAR(20)	DEFAULT 'active'	active / suspended / deactivated
`last_login_at`	TIMESTAMP	NULLABLE	
`created_at`	TIMESTAMP	DEFAULT NOW()	
`updated_at`	TIMESTAMP	DEFAULT NOW()	

Indexes: `idx_users_company`, `idx_users_email` (UNIQUE)

Table 5: `roles`

C# Model: `Role` | **Purpose:** Role definitions per company

Column	Type	Constraints	Notes

`id`	INT	PK	
`name`	VARCHAR(100)	NOT NULL	
`company_id`	INT	FK → companies.id, NULLABLE	NULL = system role
`description`	VARCHAR(500)	NULLABLE	
`is_system_role`	BOOL	DEFAULT false	Admin/CMO/PPP/Expert cannot be deleted
`color`	VARCHAR(50)	NULLABLE	role badge color
`icon`	VARCHAR(10)	NULLABLE	display icon
`created_at`	TIMESTAMP	DEFAULT NOW()	

Indexes: `idx_roles_company`

Table 6: `screens`

C# Model: `Screen` | **Purpose:** UI screen/page definitions for RBAC

Column	Type	Constraints	Notes
`id`	INT	PK	
`name`	VARCHAR(100)	NOT NULL, UNIQUE	screen identifier
`display_name`	VARCHAR(200)		human-readable name
`category`	VARCHAR(50)	NULLABLE	expert / cmo / ppp / admin / super / shared
`icon`	VARCHAR(20)	NULLABLE	emoji or icon class
`sort_order`	INT	DEFAULT 0	
`is_active`	BOOL	DEFAULT true	
`description`	VARCHAR(500)	NULLABLE	

Table 7: `role_screens`

C# Model: `RoleScreen` | **Purpose:** Role-to-screen permission mapping (RBAC matrix)

Column	Type	Constraints	Notes
`role_id`	INT	CPK, FK → roles.id	
`screen_id`	INT	CPK, FK → screens.id	
`company_id`	INT	FK → companies.id, NULLABLE	
`granted_by`	INT	FK → users.id, NULLABLE	

`granted_at`	TIMESTAMP	DEFAULT NOW()	

Key: Composite primary key (role_id + screen_id) **Indexes:** `idx_rolescreens_company`

Table 8: `user_screens`

C# Model: `UserScreen` | **Purpose:** Per-user screen permission overrides (independent of role)

Column	Type	Constraints	Notes
`user_id`	INT	CPK, FK → users.id	CASCADE on delete
`screen_id`	INT	CPK, FK → screens.id	
`company_id`	INT	FK → companies.id, NULLABLE	
`granted_by`	INT	FK → users.id, NULLABLE	SET NULL on delete
`granted_at`	TIMESTAMP	DEFAULT NOW()	

Key: Composite primary key (user_id + screen_id) **Indexes:** `idx_userscreens_company`, `idx_userscreens_user`

Table 9: `refresh_tokens`

C# Model: `RefreshToken` | **Purpose:** JWT refresh token storage for secure token rotation

Column	Type	Constraints	Notes
`id`	INT	PK	
`user_id`	INT	FK → users.id	
`token`	VARCHAR(500)	NOT NULL, UNIQUE	
`expires_at`	TIMESTAMP	NOT NULL	
`created_at`	TIMESTAMP	DEFAULT NOW()	
`revoked_at`	TIMESTAMP	NULLABLE	
`replaced_by`	VARCHAR(500)	NULLABLE	token rotation chain
`ip_address`	VARCHAR(50)	NULLABLE	
`user_agent`	VARCHAR(500)	NULLABLE	

Indexes: `uq_refresh_token` (UNIQUE), `idx_refresh_user`

DOMAIN 3 — Campaign Core (6 Tables)

Table 10: `campaign_objectives`

C# Model: `CampaignObjective` | **Purpose:** Standardized objective lookup mapped to each platform's API enum

Column	Type	Constraints	Notes

`id`	INT	PK	
`category`	VARCHAR(50)	NOT NULL	awareness / consideration / conversion
`name`	VARCHAR(100)	NOT NULL	e.g. "Brand Awareness", "Traffic"
`description`	TEXT	NULLABLE	
`icon`	VARCHAR(10)	NULLABLE	
`platform_mapping`	JSONB	NOT NULL	{facebook, tiktok, youtube, google_ads} enums
`supported_platforms`	TEXT[]	DEFAULT '{facebook,tiktok}'	
`sort_order`	INT	DEFAULT 0	
`is_active`	BOOL	DEFAULT true	

Table 11: `campaigns`

C# Model: `Campaign` | **Purpose:** Core campaign entity — full lifecycle from draft to completed

Column	Type	Constraints	Notes
`id`	INT	PK	
`company_id`	INT	FK → companies.id	
`name`	VARCHAR(200)	NULLABLE	
`objective_id`	INT	FK → campaign_objectives.id, NULLABLE	
`campaign_type`	VARCHAR(50)	DEFAULT 'standard'	standard / ab_test / dynamic
`brief`	TEXT	NULLABLE	AI strategy brief
`style_preset`	VARCHAR	NULLABLE	Cinematic / Minimalism / Cyberpunk / Vintage
`aspect_ratio`	VARCHAR	NULLABLE	1:1 / 16:9 / 9:16
`total_budget`	DECIMAL	NULLABLE	
`daily_budget`	DECIMAL	NULLABLE	
`lifetime_budget`	DECIMAL	NULLABLE	
`bid_strategy`	VARCHAR(50)	NULLABLE	lowest_cost / cost_cap / bid_cap
`currency`	VARCHAR(10)	DEFAULT 'USD'	
`start_date`	TIMESTAMP	NULLABLE	
`end_date`	TIMESTAMP	NULLABLE	

`platforms`	TEXT[]	DEFAULT '{}'	facebook / tiktok / youtube / google
`targeting`	JSONB	NULLABLE	full targeting spec
`status`	VARCHAR(30)	DEFAULT 'draft'	draft→pending_review→approved→deploying
`rejection_reason`	TEXT	NULLABLE	
`created_by`	INT	FK → users.id, SET NULL	
`approved_by`	INT	FK → users.id, SET NULL	
`deployed_by`	INT	FK → users.id, SET NULL	
`approved_at`	TIMESTAMP	NULLABLE	
`deployed_at`	TIMESTAMP	NULLABLE	
`completed_at`	TIMESTAMP	NULLABLE	
`created_at`	TIMESTAMP	DEFAULT NOW()	
`updated_at`	TIMESTAMP	DEFAULT NOW()	

Indexes: `idx_campaigns_company`, `idx_campaigns_status`, `idx_campaigns_dates`

Table 12: `ad_sets`

C# Model: `AdSet` | **Purpose:** Ad group level — targeting, budget, placements between Campaign and Ad

Column	Type	Constraints	Notes
`id`	INT	PK	
`company_id`	INT	FK → companies.id	
`campaign_id`	INT	FK → campaigns.id	
`name`	VARCHAR(200)	NOT NULL	
`status`	VARCHAR(20)	DEFAULT 'draft'	
`daily_budget`	DECIMAL	NULLABLE	
`lifetime_budget`	DECIMAL	NULLABLE	
`bid_strategy`	VARCHAR(50)	NULLABLE	
`bid_amount`	DECIMAL	NULLABLE	
`optimization_goal`	VARCHAR(50)	NULLABLE	link_clicks / impressions / conversions
`billing_event`	VARCHAR(50)	DEFAULT 'IMPRESSIONS'	

`start_time`	TIMESTAMP	NULLABLE	
`end_time`	TIMESTAMP	NULLABLE	
`targeting`	JSONB	NULLABLE	geo, age, gender, interests, audiences
`placements`	JSONB	NULLABLE	per-platform placement config
`schedule`	JSONB	NULLABLE	dayparting rules
`platform_adset_ids`	JSONB	NULLABLE	{facebook: "123", tiktok: "456"}
`created_by`	INT	FK → users.id	
`created_at`	TIMESTAMP	DEFAULT NOW()	
`updated_at`	TIMESTAMP	DEFAULT NOW()	

Indexes: `idx_adsets_campaign` , `idx_adsets_company`

Table 13: `ads`

C# Model: `Ad` | **Purpose:** Individual ad entity — specific creative + copy combination within an ad set

Column	Type	Constraints	Notes
`id`	INT	PK	
`company_id`	INT	FK → companies.id	
`ad_set_id`	INT	FK → ad_sets.id	
`name`	VARCHAR(200)	NOT NULL	
`status`	VARCHAR(20)	DEFAULT 'draft'	draft / pending_review / approved / rejected / active / paused / archived
`headline`	VARCHAR(500)	NULLABLE	
`description`	TEXT	NULLABLE	
`cta_type`	VARCHAR(50)	NULLABLE	LEARN_MORE / SHOP_NOW / SIGN_UP / BOOK_NOW / DOWNLOAD / CONTACT_US
`cta_link`	VARCHAR(1000)	NULLABLE	
`platform_ad_ids`	JSONB	NULLABLE	{facebook: "ad_123", tiktok: "ad_456"}
`review_status`	VARCHAR(20)	DEFAULT 'pending'	pending / approved / rejected
`created_by`	INT	FK → users.id, SET NULL	
`reviewed_by`	INT	FK → users.id, SET NULL	

`reviewed_at`	TIMESTAMP	NULLABLE	
`review_notes`	TEXT	NULLABLE	
`created_at`	TIMESTAMP	DEFAULT NOW()	
`updated_at`	TIMESTAMP	DEFAULT NOW()	

Indexes: `idx_ads_adset`, `idx_ads_company`, `idx_ads_status`

Table 14: `ad_creatives`

C# Model: `AdCreative` | **Purpose:** Creative assets linked to individual ads (one ad can have multiple for carousel)

Column	Type	Constraints	Notes
`id`	INT	PK	
`company_id`	INT	FK → companies.id	
`ad_id`	INT	FK → ads.id	
`creative_type`	VARCHAR(30)	NOT NULL	image / video / carousel / slideshow
`asset_url`	VARCHAR(1000)	NOT NULL	
`asset_filename`	VARCHAR(500)	NULLABLE	
`thumbnail_url`	VARCHAR(1000)	NULLABLE	
`primary_text`	TEXT	NULLABLE	main ad copy
`headline`	VARCHAR(500)	NULLABLE	
`description`	TEXT	NULLABLE	
`cta_type`	VARCHAR(50)	NULLABLE	
`cta_link`	VARCHAR(1000)	NULLABLE	
`platform_creative_ids`	JSONB	NULLABLE	{facebook: "cr_123"}
`file_size_bytes`	BIGINT	NULLABLE	
`width`	INT	NULLABLE	
`height`	INT	NULLABLE	
`duration_seconds`	INT	NULLABLE	video only
`created_at`	TIMESTAMP	DEFAULT NOW()	

Indexes: `idx_creatives_ad`, `idx_creatives_company`

Table 15: `campaign_workflow_steps`

C# Model: `CampaignWorkflowStep` | **Purpose:** Tracks campaign creation wizard progress (7 steps per campaign)

Column	Type	Constraints	Notes
`id`	INT	PK	
`campaign_id`	INT	FK → campaigns.id	
`step_name`	VARCHAR(50)	NOT NULL	objective / targeting / strategy / adset_config / creative / review / deploy
`step_order`	INT	NOT NULL	
`status`	VARCHAR(20)	DEFAULT 'not_started'	not_started / in_progress / completed / skipped
`data`	JSONB	NULLABLE	step payload snapshot
`notes`	TEXT	NULLABLE	
`completed_by`	INT	FK → users.id, NULLABLE	
`started_at`	TIMESTAMP	NULLABLE	
`completed_at`	TIMESTAMP	NULLABLE	

Indexes: `idx_workflow_campaign`, `uq_workflow_step` (campaign_id + step_name UNIQUE)

DOMAIN 4 — Workflow Queues (4 Tables)

Table 16: `cmo_queue`

C# Model: `CmoQueueItem` | **Purpose:** CMO approval queue — assets submitted by Expert, reviewed by CMO

Column	Type	Constraints	Notes
`id`	VARCHAR	PK	asset filename
`company_id`	INT	FK → companies.id	
`campaign_id`	INT	FK → campaigns.id, NULLABLE	
`ad_id`	INT	FK → ads.id, NULLABLE	
`url`	VARCHAR	NULLABLE	asset URL
`title`	VARCHAR	NULLABLE	
`type`	VARCHAR	NULLABLE	image / video
`status`	VARCHAR	DEFAULT 'pending'	pending / approved / rejected
`priority`	INT	DEFAULT 0	0=normal / 1=high / 2=urgent
`submitted_by`	INT	FK → users.id, SET NULL	
`reviewed_by`	INT	FK → users.id, SET NULL	

`reviewed_at`	TIMESTAMP	NULLABLE	
`review_notes`	TEXT	NULLABLE	
`added_at`	TIMESTAMP	DEFAULT NOW()	

Indexes: `idx_cmoqueue_company` , `idx_cmoqueue_status`

Table 17: `ppp_queue`

C# Model: `PppQueueItem` | **Purpose:** PPP dispatch queue — replaces `ppc_queue.json` file

Column	Type	Constraints	Notes
`id`	INT	PK	
`company_id`	INT	FK → companies.id	
`campaign_id`	INT	FK → campaigns.id, NULLABLE	
`ad_id`	INT	FK → ads.id, NULLABLE	
`asset_filename`	VARCHAR(500)	NOT NULL	
`asset_url`	VARCHAR(1000)	NOT NULL	
`asset_type`	VARCHAR(20)	NOT NULL	image / video
`title`	VARCHAR(500)	NULLABLE	
`status`	VARCHAR(20)	DEFAULT 'pending'	pending / selected / deploying / deployed / failed
`queue_index`	INT	DEFAULT 0	
`approved_by`	INT	FK → users.id, SET NULL	CMO who approved
`approved_at`	TIMESTAMP	NULLABLE	
`deployed_by`	INT	FK → users.id, SET NULL	PPP who deployed
`deployed_at`	TIMESTAMP	NULLABLE	
`deploy_platforms`	TEXT[]	NULLABLE	platforms deployed to
`added_at`	TIMESTAMP	DEFAULT NOW()	

Indexes: `idx_pppqueue_company` , `idx_pppqueue_status`

Table 18: `approval_comments`

C# Model: `ApprovalComment` | **Purpose:** Review feedback during approval workflow

Column	Type	Constraints	Notes

`id`	INT	PK	
`company_id`	INT	FK → companies.id	
`campaign_id`	INT	FK → campaigns.id	
`ad_id`	INT	FK → ads.id, NULLABLE	
`user_id`	INT	FK → users.id	
`comment`	TEXT	NOT NULL	
`action`	VARCHAR(30)	NOT NULL	approve / reject / request_changes / comment
`attachment_url`	VARCHAR(1000)	NULLABLE	
`created_at`	TIMESTAMP	DEFAULT NOW()	

Indexes: `idx_comments_campaign`, `idx_comments_ad`

Table 19: `brand_guidelines`

C# Model: `BrandGuideline` | **Purpose:** Brand identity configuration per company

Column	Type	Notes
`id`	INT PK	
`company_id`	INT FK	
`brand_label`	VARCHAR	
`tone`	VARCHAR	
`language`	VARCHAR	
`description`	VARCHAR	
`tagline`	VARCHAR(500)	
`logo_url`	VARCHAR(500)	
`whitelist`	TEXT	approved words
`blacklist`	TEXT	banned words
`voice_examples`	TEXT	example copy
`do_list`	TEXT	brand dos
`dont_list`	TEXT	brand don'ts
`typography`	JSONB	font config
`palette`	TEXT[]	color array

created_by	INT FK	SET NULL
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

Indexes: `idx_guidelines_company`

DOMAIN 5 — Analytics (2 Tables)

Table 20: `ad_metrics`

C# Model: `AdMetric` | **Purpose:** Daily performance metrics fetched from each ad platform

Column	Type	Notes
id	INT PK	
company_id	INT FK	
campaign_id	INT FK	
ad_set_id	INT FK NULLABLE	
ad_id	INT FK NULLABLE	
platform	VARCHAR(50)	facebook / tiktok / youtube
date	DATE	
impressions	BIGINT	DEFAULT 0
reach	BIGINT	DEFAULT 0
clicks	BIGINT	DEFAULT 0
ctr	DECIMAL(8,4)	click-through rate %
cpc	DECIMAL(10,4)	cost per click
cpm	DECIMAL(10,4)	cost per 1000 impressions
spend	DECIMAL(12,2)	
conversions	INT	
conversion_value	DECIMAL(12,2)	
roas	DECIMAL(10,4)	return on ad spend
frequency	DECIMAL(8,2)	
video_views	BIGINT NULLABLE	
video_completions	BIGINT NULLABLE	
leads	INT NULLABLE	

`app_installs`	INT NULLABLE	
`fetched_at`	TIMESTAMP	

Indexes: `idx_metrics_company` , `idx_metrics_campaign` , `idx_metrics_date` **Unique:** `uq_metrics_daily` (campaign_id + ad_set_id + ad_id + platform + date)

Table 21: `deployment_logs`

C# Model: `DeploymentLog` | **Purpose:** Audit trail for every API call made during deployment

Column	Type	Notes
`id`	INT PK	
`company_id`	INT FK	
`campaign_id`	INT FK	
`ad_set_id`	INT FK NULLABLE	
`ad_id`	INT FK NULLABLE	
`platform`	VARCHAR(50)	
`action`	VARCHAR(50)	upload_media / create_campaign / create_adset / create_creative / create_ad / pause / resume / delete
`platform_resource_id`	VARCHAR(200) NULLABLE	ID returned by platform API
`status`	VARCHAR(20)	success / failed / pending / simulated
`request_payload`	JSONB NULLABLE	what was sent
`response_payload`	JSONB NULLABLE	what was received
`error_message`	TEXT NULLABLE	
`duration_ms`	INT NULLABLE	
`executed_by`	INT FK	RESTRICT delete
`executed_at`	TIMESTAMP	

Indexes: `idx_deplogs_company` , `idx_deplogs_campaign` , `idx_deplogs_date`

DOMAIN 6 — Features (4 Tables)

Table 22: `ab_tests`

C# Model: `AbTest` | **Purpose:** A/B testing framework with statistical significance tracking

Column	Type	Notes

`id`	INT PK	
`company_id`	INT FK	
`campaign_id`	INT FK	
`name`	VARCHAR(200)	
`variant_a_ad_id`	INT FK → ads.id	RESTRICT delete
`variant_b_ad_id`	INT FK → ads.id	RESTRICT delete
`metric`	VARCHAR(50)	ctr / cpc / conversion_rate / roas
`traffic_split`	INT	DEFAULT 50 (% for variant A)
`status`	VARCHAR(20)	draft / running / completed / cancelled
`winner`	VARCHAR(5) NULLABLE	A / B / inconclusive
`confidence_level`	DECIMAL(5,2) NULLABLE	statistical significance %
`variant_a_result`	DECIMAL(12,4) NULLABLE	
`variant_b_result`	DECIMAL(12,4) NULLABLE	
`created_by`	INT FK NULLABLE	
`started_at`	TIMESTAMP NULLABLE	
`ended_at`	TIMESTAMP NULLABLE	
`created_at`	TIMESTAMP	

Indexes: `idx_abtests_company`, `idx_abtests_campaign`

Table 23: `budget_allocations`

C# Model: `BudgetAllocation` | **Purpose:** Budget planning & spend tracking per platform per period

Column	Type	Notes
`id`	INT PK	
`company_id`	INT FK	
`period_type`	VARCHAR(20)	monthly / quarterly / yearly
`period_start`	DATE	
`period_end`	DATE	
`total_budget`	DECIMAL(14,2)	
`facebook_allocation`	DECIMAL(14,2)	DEFAULT 0
`tiktok_allocation`	DECIMAL(14,2)	DEFAULT 0

`youtube_allocation`	DECIMAL(14,2)	DEFAULT 0
`google_allocation`	DECIMAL(14,2)	DEFAULT 0
`spent_to_date`	DECIMAL(14,2)	DEFAULT 0
`status`	VARCHAR(20)	active / closed / exceeded
`created_by`	INT FK NULLABLE	
`created_at`	TIMESTAMP	
`updated_at`	TIMESTAMP	

Indexes: `idx_budgets_company` , `idx_budgets_period`

Table 24: `campaign_templates`

C# Model: `CampaignTemplate` | **Purpose:** Reusable campaign configuration templates

Column	Type	Notes
`id`	INT PK	
`company_id`	INT FK NULLABLE	NULL = global (Super Admin template)
`name`	VARCHAR(200)	
`description`	TEXT NULLABLE	
`objective_id`	INT FK NULLABLE	
`targeting`	JSONB NULLABLE	
`budget_config`	JSONB NULLABLE	{daily_budget, bid_strategy}
`creative_specs`	JSONB NULLABLE	{style_preset, aspect_ratio, formats}
`platforms`	TEXT[] NULLABLE	
`is_global`	BOOL	DEFAULT false
`created_by`	INT FK NULLABLE	
`use_count`	INT	DEFAULT 0
`created_at`	TIMESTAMP	
`updated_at`	TIMESTAMP	

Indexes: `idx_templates_company`

Table 25: `audience_templates`

C# Model: `AudienceTemplate` | **Purpose:** Saved reusable audience targeting configurations

Column	Type	Notes

`id`	INT PK	
`company_id`	INT FK	
`name`	VARCHAR(200)	
`description`	TEXT NULLABLE	
`targeting`	JSONB NOT NULL	full targeting spec
`estimated_size`	BIGINT NULLABLE	estimated audience size
`created_by`	INT FK NULLABLE	
`use_count`	INT	DEFAULT 0
`created_at`	TIMESTAMP	
`updated_at`	TIMESTAMP	

Indexes: `idx_audiences_company`

DOMAIN 7 — System (3 Tables)

Table 26: `notifications`

C# Model: `Notification` | **Purpose:** In-app notification system for all workflow events

Column	Type	Notes
`id`	INT PK	
`company_id`	INT FK	
`user_id`	INT FK	recipient
`type`	VARCHAR(50)	campaign_submitted / campaign_approved / deploy_success / budget_alert / etc.
`title`	VARCHAR(200)	
`message`	TEXT	
`resource_type`	VARCHAR(50) NULLABLE	campaign / ad / deployment / user / company
`resource_id`	INT NULLABLE	
`action_url`	VARCHAR(500) NULLABLE	deep link
`is_read`	BOOL	DEFAULT false
`read_at`	TIMESTAMP NULLABLE	
`created_at`	TIMESTAMP	

Indexes: `idx_notif_user` (user_id + is_read), `idx_notif_company`

Table 27: `activity_logs`

C# Model: `ActivityLog` | **Purpose:** Full audit trail — every significant action for compliance

Column	Type	Notes
`id`	INT PK	
`company_id`	INT FK NULLABLE	NULL = super admin action
`user_id`	INT FK	
`action`	VARCHAR(50)	login / logout / created / updated / deleted / approved / deployed / invited / role_changed
`resource_type`	VARCHAR(50)	user / campaign / ad / company / role / deployment
`resource_id`	VARCHAR(50) NULLABLE	
`description`	TEXT NULLABLE	
`details`	JSONB NULLABLE	before/after snapshot
`ip_address`	VARCHAR(50) NULLABLE	
`user_agent`	VARCHAR(500) NULLABLE	
`created_at`	TIMESTAMP	

Indexes: `idx_activity_company` (company_id + created_at), `idx_activity_user` , `idx_activity_date`

Table 28: `invitations`

C# Model: `Invitation` | **Purpose:** Team member invitation via email with pre-assigned role

Column	Type	Notes
`id`	INT PK	
`company_id`	INT FK	
`email`	VARCHAR(200) NOT NULL	
`role_id`	INT FK → roles.id	
`invited_by`	INT FK → users.id	
`token`	VARCHAR(200) NOT NULL UNIQUE	

`status`	VARCHAR(20)	DEFAULT 'pending' — pending / accepted / expired / revoked
`accepted_at`	TIMESTAMP NULLABLE	
`expires_at`	TIMESTAMP NOT NULL	
`created_at`	TIMESTAMP	

Indexes: `uq_invitation_token` (UNIQUE), `idx_invitation_company`, `idx_invitation_email`

DOMAIN 8 — Assets (1 Table)

Table 29: `asset_library`

C# Model: `AssetLibraryItem` | **Purpose:** Database-tracked asset management with metadata and tagging

Column	Type	Notes
`id`	INT PK	
`company_id`	INT FK	
`filename`	VARCHAR(500) NOT NULL	
`original_name`	VARCHAR(500) NULLABLE	
`file_path`	VARCHAR(1000) NOT NULL	
`file_url`	VARCHAR(1000) NOT NULL	
`file_type`	VARCHAR(20) NOT NULL	image / video
`mime_type`	VARCHAR(100) NULLABLE	
`file_size_bytes`	BIGINT NULLABLE	
`width`	INT NULLABLE	
`height`	INT NULLABLE	
`duration_seconds`	INT NULLABLE	video only
`folder`	VARCHAR(50)	DEFAULT 'assets' — assets / library
`tags`	TEXT[] NULLABLE	
`status`	VARCHAR(20)	DEFAULT 'active' — active / archived / deleted
`uploaded_by`	INT FK	
`uploaded_at`	TIMESTAMP	

Indexes: `idx_assetlib_company`, `idx_assetlib_folder` (company_id + folder)

---

3. API Endpoints — Complete Reference

Total Implemented: 95+ endpoints

Auth (3)

Method	Endpoint	Description
POST	`/api/auth/register`	Create user account
POST	`/api/auth/login`	Authenticate → JWT + screens list
POST	`/api/auth/refresh`	Refresh JWT token
POST	`/api/onboard/company`	Company + admin onboarding wizard

Super Admin (8)

Method	Endpoint	Description
GET	`/api/super-admin/dashboard`	Platform-wide stats
GET	`/api/super-admin/companies`	List all companies
POST	`/api/super-admin/companies`	Create company
GET	`/api/super-admin/companies/{id}`	Get company details
PUT	`/api/super-admin/companies/{id}`	Update company
DELETE	`/api/super-admin/companies/{id}`	Delete company
GET	`/api/super-admin/companies/{id}/users`	Company users
GET	`/api/super-admin/companies/{id}/campaigns`	Company campaigns
GET	`/api/super-admin/audit-log`	Full audit log

RBAC (12)

Method	Endpoint	Description
GET	`/api/rbac/roles`	List roles
POST	`/api/rbac/roles`	Create role
PUT	`/api/rbac/roles/{id}`	Update role
DELETE	`/api/rbac/roles/{id}`	Delete role
GET	`/api/rbac/screens`	List screens (filtered by role)
GET	`/api/rbac/users`	List users
DELETE	`/api/rbac/users/{id}`	Revoke user
PATCH	`/api/rbac/users/{id}/activate`	Re-activate user

DELETE	`/api/rbac/users/{id}/permanent`	Permanently delete user
GET	`/api/rbac/role-permissions/{roleId}`	Get role screen list
POST	`/api/rbac/role-permissions`	Save role-screen matrix
GET	`/api/rbac/user-permissions/{userId}`	Get user-level overrides
POST	`/api/rbac/user-permissions`	Save user-level overrides
POST	`/api/rbac/seed`	Seed default roles/screens

Campaigns (13)

Method	Endpoint	Description
GET	`/api/campaigns`	List campaigns (filter by status)
POST	`/api/campaigns`	Create campaign
GET	`/api/campaigns/{id}`	Get campaign
PUT	`/api/campaigns/{id}`	Update campaign
DELETE	`/api/campaigns/{id}`	Delete campaign
POST	`/api/campaigns/{id}/submit`	Submit for review
POST	`/api/campaigns/{id}/approve`	Approve campaign
POST	`/api/campaigns/{id}/reject`	Reject campaign
POST	`/api/campaigns/{id}/pause`	Pause campaign
POST	`/api/campaigns/{id}/resume`	Resume campaign
GET	`/api/campaigns/{id}/workflow`	Get workflow steps
PUT	`/api/campaigns/{id}/workflow/{stepName}`	Update workflow step
POST	`/api/campaigns/{id}/comments`	Add approval comment
GET	`/api/campaigns/{id}/comments`	List comments
GET	`/api/campaign-objectives`	List all objectives

Ad Sets & Ads (11)

Method	Endpoint	Description
POST	`/api/campaigns/{id}/adsets`	Create ad set
GET	`/api/campaigns/{id}/adsets`	List ad sets
GET	`/api/campaigns/{id}/adsets/{id}`	Get ad set
PUT	`/api/campaigns/{id}/adsets/{id}`	Update ad set

DELETE	`/api/campaigns/{id}/adsets/{id}`	Delete ad set
POST	`/api/adsets/{id}/ads`	Create ad
GET	`/api/adsets/{id}/ads`	List ads
GET	`/api/adsets/{id}/ads/{id}`	Get ad
PUT	`/api/adsets/{id}/ads/{id}`	Update ad
DELETE	`/api/adsets/{id}/ads/{id}`	Delete ad
POST	`/api/adsets/{id}/ads/{id}/review`	Review/approve ad

Queues (6)

Method	Endpoint	Description
GET	`/api/cmo/queue`	CMO approval queue
POST	`/api/cmo/queue`	Update CMO queue item
GET	`/api/ppp/queue`	PPP dispatch queue (DB-backed)
POST	`/api/ppp/queue`	Update PPP queue item
GET	`/api/ppc/queue`	Legacy PPC queue (file-based)
POST	`/api/ppc/queue`	Legacy PPC queue update

Assets (8)

Method	Endpoint	Description
GET	`/api/assets`	List /Assets working files
POST	`/api/assets/save-url`	Download image from URL to /Assets
DELETE	`/api/assets/{filename}`	Delete asset
POST	`/api/assets/approve`	Copy to /Assets Library
GET	`/api/assets-library`	List /Assets Library files
GET	`/api/assets/company/{companyId}`	Company-scoped assets
GET	`/api/assets-library/company/{companyId}`	Company-scoped library
POST	`/api/compliance/check`	Brand compliance check on asset

Deploy (3)

Method	Endpoint	Description
POST	`/api/deploy/facebook`	Deploy to Facebook (Campaign→AdSet→Creative→Ad)
POST	`/api/deploy/tiktok-adgroup`	Deploy to TikTok (AdGroup→Creative)

POST	`/api/deploy/unified`	Unified multi-platform deployment orchestrator

AI / Gemini (2)

Method	Endpoint	Description
POST	`/api/gemini/questions`	Generate 5 Level-1 strategy questions
POST	`/api/gemini/follow-up`	Generate 5 Level-2 deep-dive questions

Analytics (6)

Method	Endpoint	Description
GET	`/api/analytics/overview`	Platform analytics overview
GET	`/api/analytics/campaigns/{id}/metrics`	Campaign metrics by platform/date
GET	`/api/analytics/platforms`	Cross-platform comparison
GET	`/api/analytics/top-performers`	Top performing campaigns/ads
GET	`/api/deployment-logs`	Deployment audit logs
GET	`/api/activity-logs`	User activity logs

A/B Tests (5)

Method	Endpoint	Description
POST	`/api/ab-tests`	Create A/B test
GET	`/api/ab-tests`	List A/B tests
GET	`/api/ab-tests/{id}/results`	Get test results
POST	`/api/ab-tests/{id}/end`	End A/B test
POST	`/api/ab-tests/{id}/optimize`	Apply winner optimization

Budgets (3)

Method	Endpoint	Description
POST	`/api/budgets`	Create budget allocation
GET	`/api/budgets`	List budget allocations
GET	`/api/budgets/current`	Get current period budget

Templates & Audiences (5)

Method	Endpoint	Description
POST	`/api/templates`	Create campaign template

GET	`/api/templates`	List campaign templates
DELETE	`/api/templates/{id}`	Delete template
POST	`/api/audience-templates`	Create audience template
GET	`/api/audience-templates`	List audience templates

Notifications & Invitations (7)

Method	Endpoint	Description
GET	`/api/notifications`	List notifications for user
PUT	`/api/notifications/{id}/read`	Mark as read
POST	`/api/notifications/read-all`	Mark all as read
POST	`/api/invitations`	Create invitation
GET	`/api/invitations`	List invitations
POST	`/api/invitations/{token}/accept`	Accept invitation
DELETE	`/api/invitations/{id}`	Revoke invitation

Ad Accounts & Settings (6)

Method	Endpoint	Description
GET	`/api/ad-accounts`	List company ad accounts
POST	`/api/ad-accounts`	Add ad account
PUT	`/api/ad-accounts/{id}`	Update ad account
DELETE	`/api/ad-accounts/{id}`	Remove ad account
GET	`/api/company-settings`	Get company settings
PUT	`/api/company-settings`	Update company settings

Brand Guidelines (2)

Method	Endpoint	Description
GET	`/api/guidelines`	Fetch brand guideline
POST	`/api/guidelines`	Save brand guideline

Utilities (4)

Method	Endpoint	Description
GET	`/api/search`	Global search across campaigns/assets

GET	`/api/test/facebook-connection`	Test Facebook API connection
POST	`/api/seed/dummy-data`	Seed dummy data for testing
POST	`/api/rbac/seed`	Seed roles, screens, super admin

---

4. Database Indexes & Constraints

Unique Constraints (8)

Index Name	Table	Columns
`uq_companies_slug`	companies	slug
`uq_settings_company`	company_settings	company_id
`uq_refresh_token`	refresh_tokens	token
`uq_invitation_token`	invitations	token
`idx_users_email`	users	email
`uq_adaccounts_platform`	company_ad_accounts	company_id + platform + account_id
`uq_workflow_step`	campaign_workflow_steps	campaign_id + step_name
`uq_metrics_daily`	ad_metrics	campaign_id + ad_set_id + ad_id + platform + date

Composite Primary Keys (2)

Table	Keys
`role_screens`	role_id + screen_id
`user_screens`	user_id + screen_id

Foreign Key Delete Behaviors

Relationship	Behavior
campaign → created_by / approved_by / deployed_by	SET NULL
ad → created_by / reviewed_by	SET NULL
ab_test → variant_a / variant_b	RESTRICT
cmo_queue → submitted_by / reviewed_by	SET NULL
ppp_queue → approved_by / deployed_by	SET NULL
role_screen → granted_by	SET NULL
user_screen → user	CASCADE
user_screen → granted_by	SET NULL

brand_guideline → created_by	SET NULL
deployment_log → executed_by	RESTRICT
company → company_settings	CASCADE

---

5. Summary Counts

```
╔═══════════════════════════════════════════════════╗
│              IMPLEMENTATION SUMMARY                  │
╟═══════════════════════════════════════════════════╢
│  Frontend Screens         36 screens                 │
│  ── Super Admin screens    4                         │
│  ── Admin screens          8                         │
│  ── CMO screens            8                         │
│  ── Expert screens         9                         │
│  ── PPP screens            6                         │
│  ── Shared screens         1 (Notifications)         │
├──────────────────────────────────────────────────┤
│  Database Tables          29 tables                  │
│  ── Multi-Tenancy          3                         │
│  ── Auth & RBAC            6                         │
│  ── Campaign Core          6                         │
│  ── Workflow Queues        4                         │
│  ── Analytics              2                         │
│  ── Features               4                         │
│  ── System                 3                         │
│  ── Assets                 1                         │
├──────────────────────────────────────────────────┤
│  API Endpoints            95+ endpoints              │
│  ── Auth & Onboard         4                         │
│  ── Super Admin            9                         │
│  ── RBAC                  14                         │
│  ── Campaigns             15                         │
│  ── Ad Sets & Ads         11                         │
│  ── Queues                 6                         │
│  ── Assets                 8                         │
│  ── Deploy                 3                         │
│  ── AI / Gemini            2                         │
│  ── Analytics              6                         │
│  ── A/B Tests              5                         │
│  ── Budgets                3                         │
│  ── Templates              5                         │
│  ── Notifications          7                         │
│  ── Ad Accounts            6                         │
│  ── Utilities              4                         │
├──────────────────────────────────────────────────┤
│  DB Indexes               35+ indexes                │
│  DB Unique Constraints     8                         │
│  DB Foreign Keys          50+                        │
```

```
│  DB Composite PKs          2                          │
╙──────────────────────────────────────────────────────╜
```

Document: database_and_screens.md Source: `backend/Models.cs` · `backend/AppDbContext.cs` · `backend/Program.cs` · `frontend/src/main.js` All entries verified from actual project code